

Physical Chemistry (I)

Lecture 05



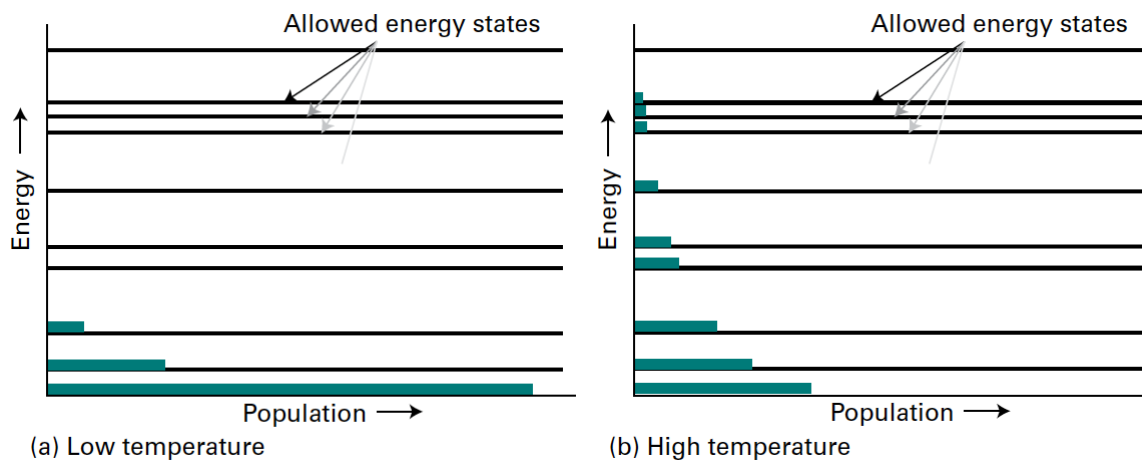
In the Last Class

- Boltzmann distribution
 - Review: probabilities
- Maxwell-Boltzmann distribution: $E \rightarrow \vec{v} \rightarrow v$
 - characteristic speeds
- Collisions
 - Molecule-molecule collisions: chemical kinetics
 - Molecule-wall collisions: pressure

Boltzmann Distribution

$$\text{Probability} \propto e^{-E/k_B T}$$

(at thermal equilibrium)



Maxwell-Boltzmann Distribution

- Velocity distribution:

$$f(v_x, v_y, v_z) = \left(\frac{M}{2\pi RT} \right)^{3/2} e^{-M(v_x^2 + v_y^2 + v_z^2)/2RT}$$

- Speed distribution (Maxwell-Boltzmann):

$$f(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Characteristic Speeds: KMT

1. Mean speed

$$\langle v \rangle = \left(\frac{8RT}{\pi M} \right)^{1/2}$$

2. Root-mean-square speed

$$v_{\text{rms}} = \left(\frac{3RT}{M} \right)^{1/2}$$

3. Most probable speed

$$v_{\text{mp}} = \left(\frac{2RT}{M} \right)^{1/2}$$

Mean Relative Speed

For two molecules of different types (A and B),

$$v_{\text{rel}} = \left(\frac{8k_{\text{B}}T}{\pi\mu} \right)^{1/2}$$

where reduced mass $\mu = m_{\text{A}}m_{\text{B}}/(m_{\text{A}} + m_{\text{B}})$

Molecule-Molecule Collisions

- Collision frequency

$$z = \sigma v_{\text{rel}} \frac{n N_A}{V} = \sigma v_{\text{rel}} \frac{p}{k_B T}$$

- Mean free path

$$\lambda = \frac{v_{\text{rel}}}{z} = \frac{k_B T}{\sigma p}$$

- Collision density for a two-component system

$$Z_{AB} = \sigma v_{\text{rel}} N_A^2 \frac{n_A}{V} \frac{n_B}{V} \propto [A][B]$$

Molecule-Wall Collisions

- Pressure

$$p = \frac{nM\langle v^2 \rangle}{3V}$$

p - T Relationship

$$pV = \frac{1}{3} nM \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\text{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$pV = nRT$$

Internal Energy, U

“The total energy of a system”

For a perfect gas,

$$U = \sum_{i=1}^N E_{k,i} = \sum_{i=1}^N \frac{1}{2} m v_i^2 = \frac{1}{2} m \sum_{i=1}^N v_i^2$$

On the other hand,

$$\langle v^2 \rangle = \frac{1}{N_A n} \sum_{i=1}^N v_i^2$$

$$\sum_{i=1}^N v_i^2 = N_A n \langle v^2 \rangle$$

U of Perfect Gas

Substitution leads to

$$U = \frac{1}{2} m N_A n \langle v^2 \rangle = \frac{1}{2} M n \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\text{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$U = \frac{3}{2} nRT$$

Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each quadratic contribution
to the internal energy is $(1/2)k_B T$.

(Quadratic contribution: $E \propto p_x^2$ or $E \propto x^2$)

Monatomic Perfect Gas

- Potential energy = 0
- Kinetic energy: translational energy

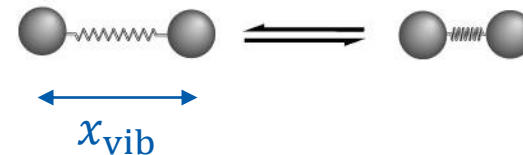
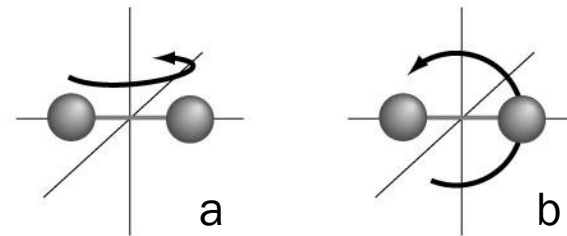
$$E_{\text{k,tr}} = \frac{p^2}{2m} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

→ 3 quadratic terms per molecule

$$U = (nN_A) \times \left(\frac{3}{2} k_B T \right) = \frac{3}{2} n N_A k_B T = \frac{3}{2} n R T$$

Diatomic Perfect Gas

- Potential energy = 0
- Kinetic energy
 - translation
 - rotation: two possible motions
- vibration: one possible motion



Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule

$$E_{k,\text{tr}} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

- Rotational energy: 2 quadratic terms per molecule

$$E_{k,\text{rot}} = \frac{p_a^2}{2m} + \frac{p_b^2}{2m}$$

- Vibrational energy: 2 quadratic terms per molecule

$$E_{k,\text{vib}} = \frac{p_{\text{vib}}^2}{2m} + \frac{1}{2} k x_{\text{vib}}^2$$

Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule
- Rotational energy: 2 quadratic terms per molecule
- Vibrational energy: 2 quadratic terms per molecule

Total: 7 quadratic terms per molecule

From the equipartition theorem,

$$U = (nN_A) \times \left(\frac{7}{2} k_B T \right) = \frac{7}{2} n N_A k_B T = \frac{7}{2} n R T$$

Polyatomic Perfect Gas

- Degree of freedom f : number of variables that may vary independently

$$f = 3a \text{ (} a: \text{ number of atoms)}$$

$$\text{CO}_2: f = 3 \times 3 = 9$$

$$\text{H}_2\text{O}: f = 3 \times 3 = 9$$

$$\text{NH}_3: f = 3 \times 4 = 12$$

Polyatomic Perfect Gas

- Linear molecules (e.g. CO₂)
 - Translation: 3 independent motions → 3 quadratic
 - Rotation: 2 independent motions → 2 quadratic
 - Vibration: $(f - 5)$ independent motions → $2(f - 5)$ quadratic

$$U = (2f - 5)nRT/2$$

- Nonlinear molecules (e.g. H₂O)
 - Translation: 3 independent motions → 3 quadratic
 - Rotation: 3 independent motions → 3 quadratic
 - Vibration: $(f - 6)$ independent motions → $2(f - 6)$ quadratic

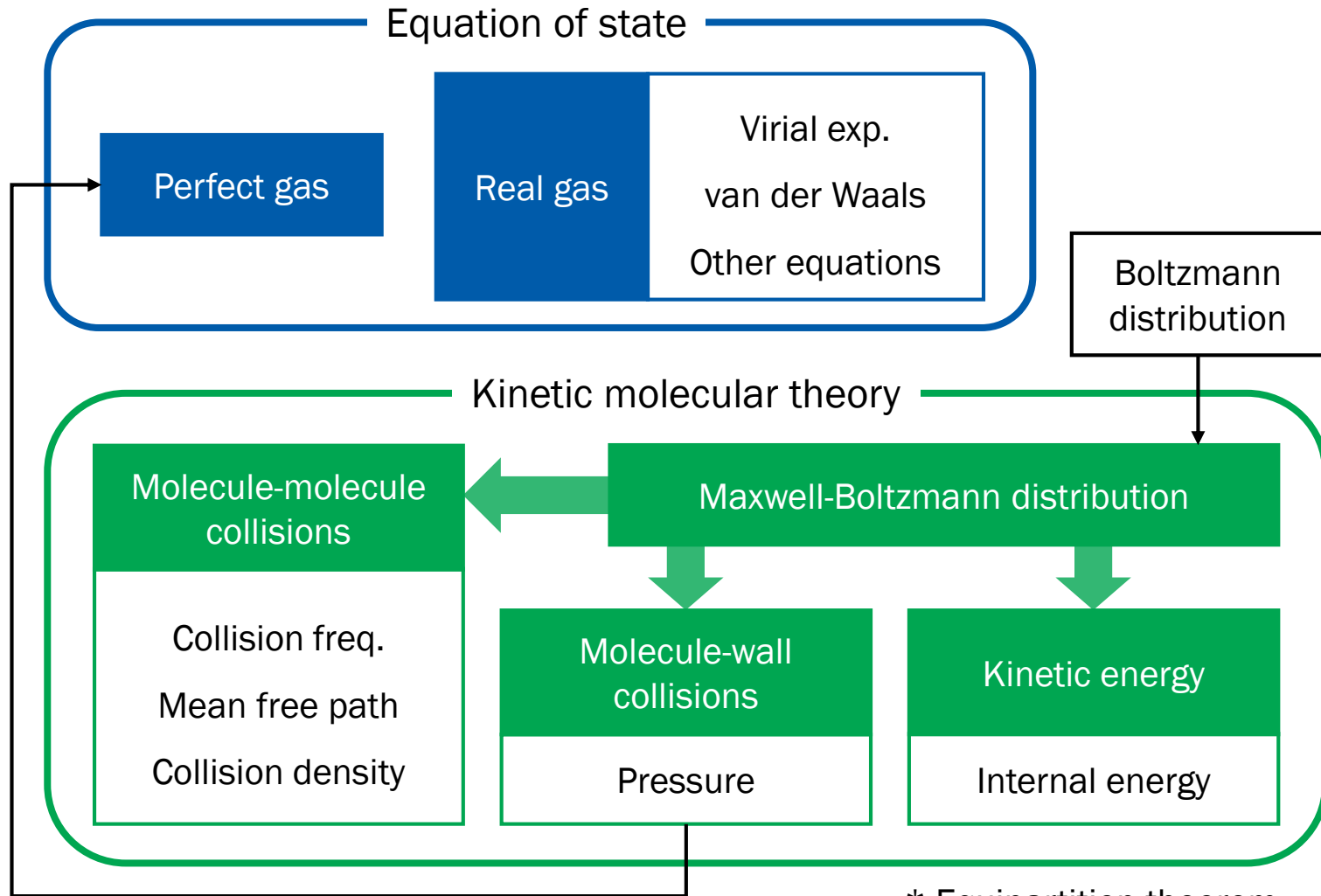
$$U = (2f - 6)nRT/2$$

U of Perfect Gas

The internal energy of a perfect gas is independent of the volume it occupies.

- No molecule-molecule interactions in a perfect gas
- The molecule-molecule distance has no effect on U

Summary



* Equipartition theorem

3. The First Law

Chapter Overview

- First law of thermodynamics
 - Work and heat
 - Differential form of internal energy
- Enthalpy
 - Definition and properties of enthalpy
 - Differential form
 - Adiabatic change
- Thermochemistry



Internal Energy, U

“The total energy of a system”

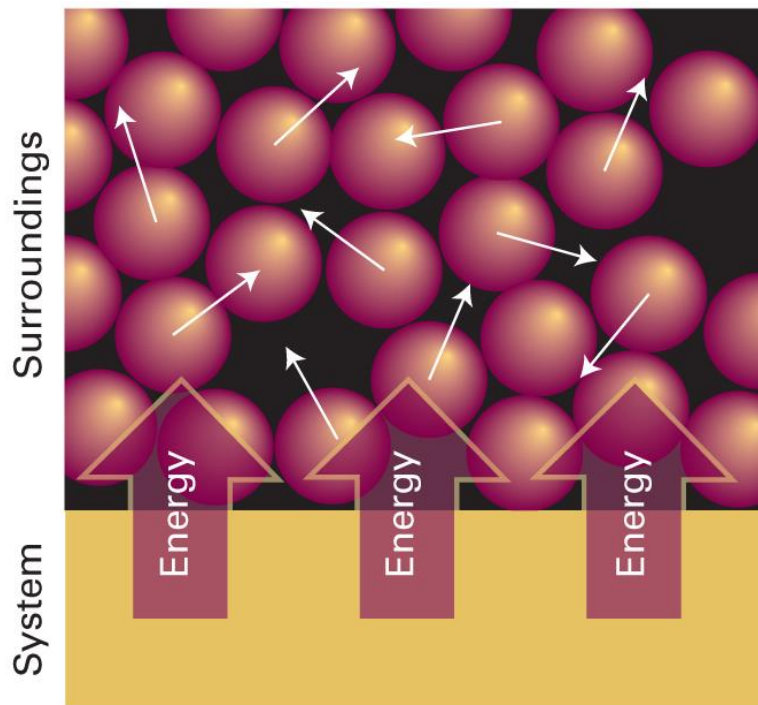
- Perfect gas (monatomic): $U = (3/2)nRT$
- Molar internal energy

$$U_m = U/n$$

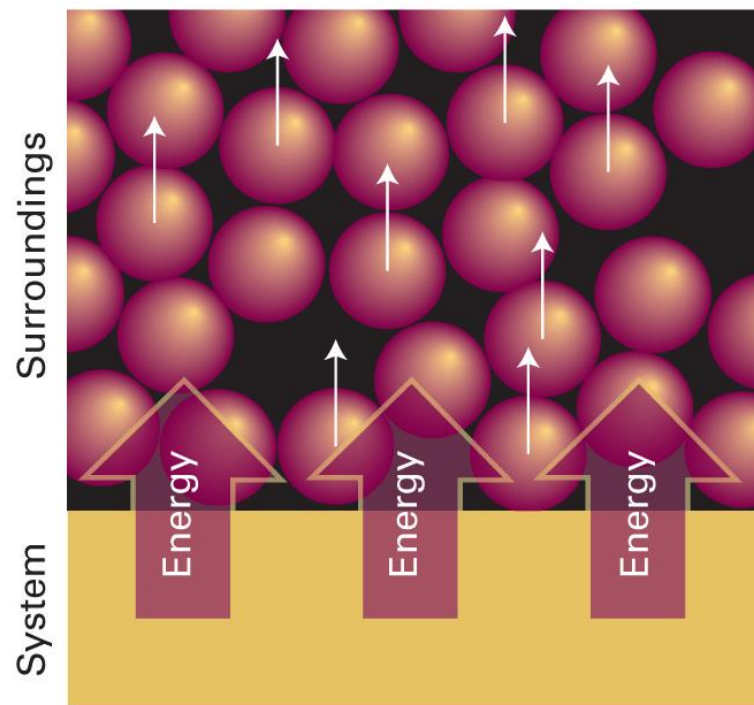
For a perfect gas, $U_m = (3/2)RT$

How can we change the internal energy?

Two Possible Ways



“Heat”



“Work”

First Law of Thermodynamics

“The internal energy of an isolated system is constant.”

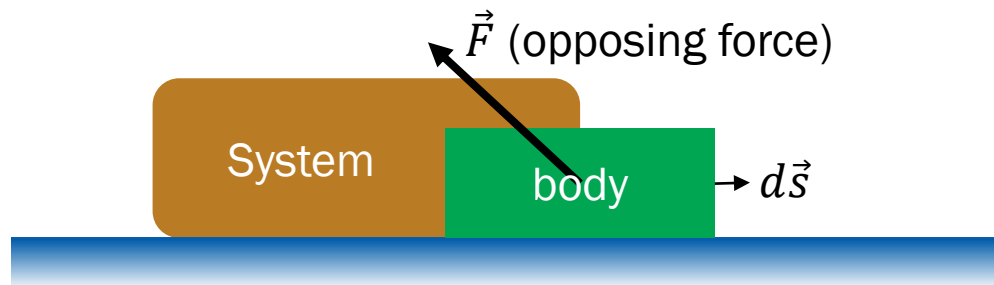
- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$dU = \delta q + \delta w$$

↑ ↑ ↑
small change of U small q small w

Work



- Work done on the body

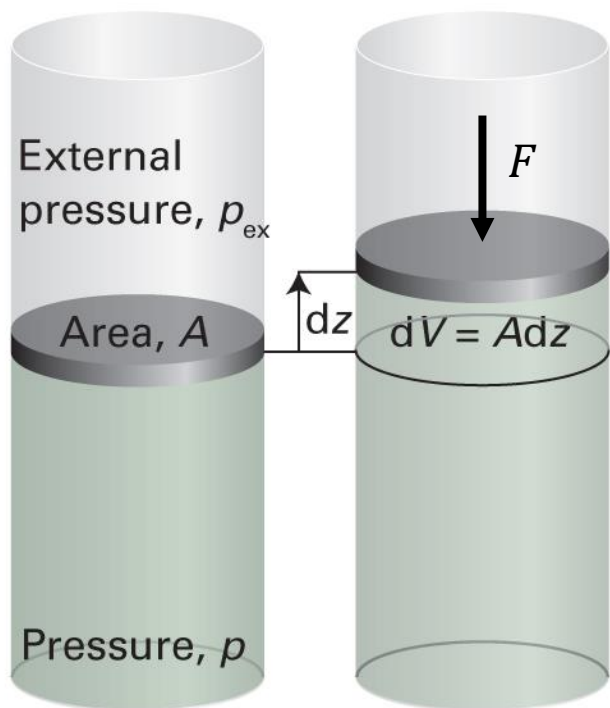
$$\delta w_{\text{body}} = -\vec{F} \cdot d\vec{s}$$

- Work done on the system

$$\delta w = \vec{F} \cdot d\vec{s}$$

where $\vec{F} \cdot d\vec{s} = F_x dx + F_y dy + F_z dz$

Expansion Work



$$\delta w = \vec{F} \cdot d\vec{s} = F_x dx + F_y dy + F_z dz$$

$$\text{As } dx = dy = 0,$$

$$\delta w = -|F|dz$$

$$\text{Since } p_{\text{ex}} = |F|/A,$$

$$\delta w = -p_{\text{ex}}Adz = -p_{\text{ex}}dV$$

Other Types of Work

$$\delta w = (\text{intensive factor}) \times (\text{extensive factor})$$

Type of work	dw	Comments	Units [†]
Expansion	$-p_{\text{ex}}dV$	p_{ex} is the external pressure	Pa
		dV is the change in volume	m^3
Surface expansion	$\gamma d\sigma$	γ is the surface tension	N m^{-1}
		$d\sigma$ is the change in area	m^2
Extension	$f dl$	f is the tension	N
		dl is the change in length	m
Electrical	ϕdQ	ϕ is the electric potential	V
		dQ is the change in charge	C
	$Q d\phi$	$d\phi$ is the potential difference	V
		Q is the charge transferred	C

Calculation of Expansion Work

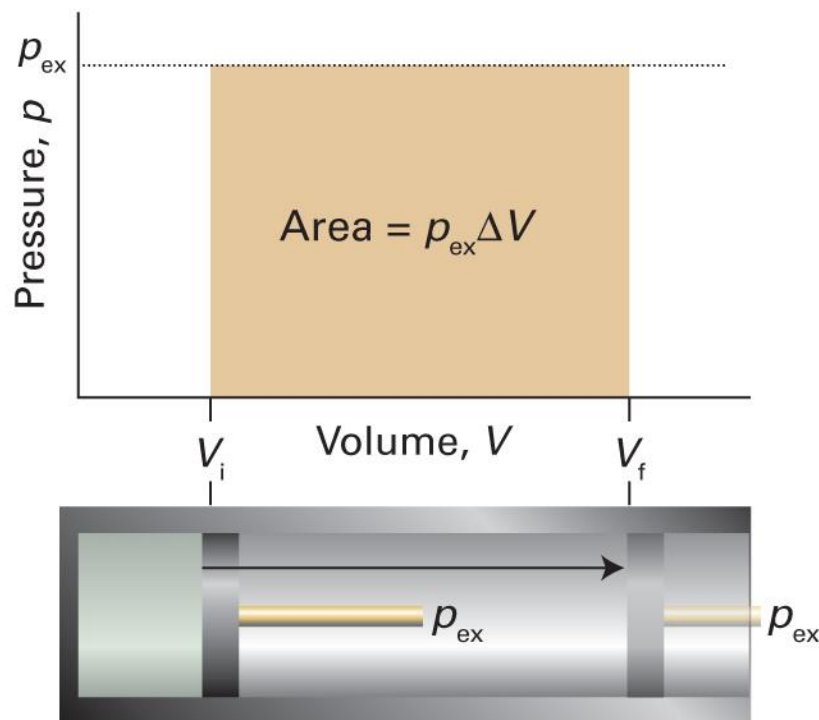
$$\delta w = -p_{\text{ex}} dV$$

- Total work for a change from **state i** to **state f**

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\text{ex}} dV) = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

1. Expansion against constant external pressure
2. Reversible expansion

Constant External Pressure



$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

$$w = -p_{\text{ex}} \int_{V_i}^{V_f} dV$$

$$w = -p_{\text{ex}}(V_f - V_i)$$

For $\Delta V = V_f - V_i$,

$$w = -p_{\text{ex}} \Delta V$$

Free Expansion

$$w = -p_{\text{ex}}\Delta V$$

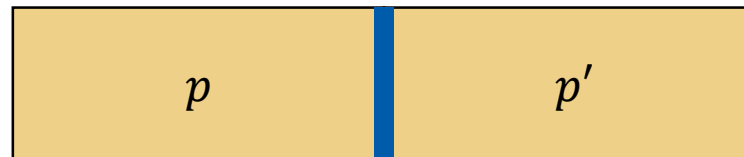
$$\text{If } p_{\text{ex}} = 0, w = 0$$

(regardless of the value of ΔV)

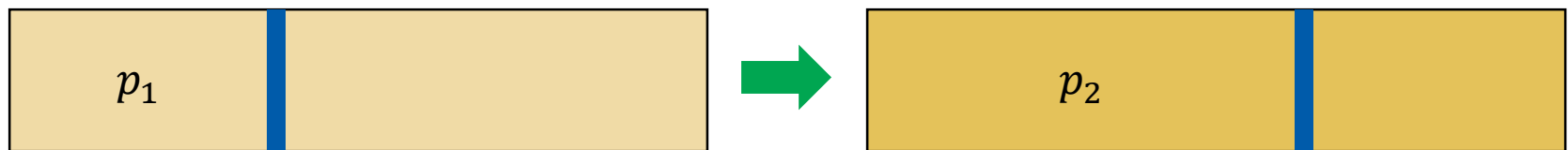
e.g. a gas expanding into a vacuum

Reversible Expansion

- Reversible change: a change that can be reversed by an **infinitesimal** modification of a variable
- The system is **at equilibrium** at any time during the change



$p = p'$ always



Reversible Expansion Work

$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

For the reversible expansion, $p = p_{\text{ex}}$ always.

$$w = - \int_{V_i}^{V_f} p dV$$

Case of Perfect Gas

$$w = - \int_{V_i}^{V_f} p \, dV$$

For the reversible expansion of a perfect gas,

$$pV = nRT$$

$$p = nRT/V$$

Hence,

$$w = - \int_{V_i}^{V_f} p \, dV = - \int_{V_i}^{V_f} \frac{nRT}{V} \, dV = -nR \int_{V_i}^{V_f} \frac{T}{V} \, dV$$

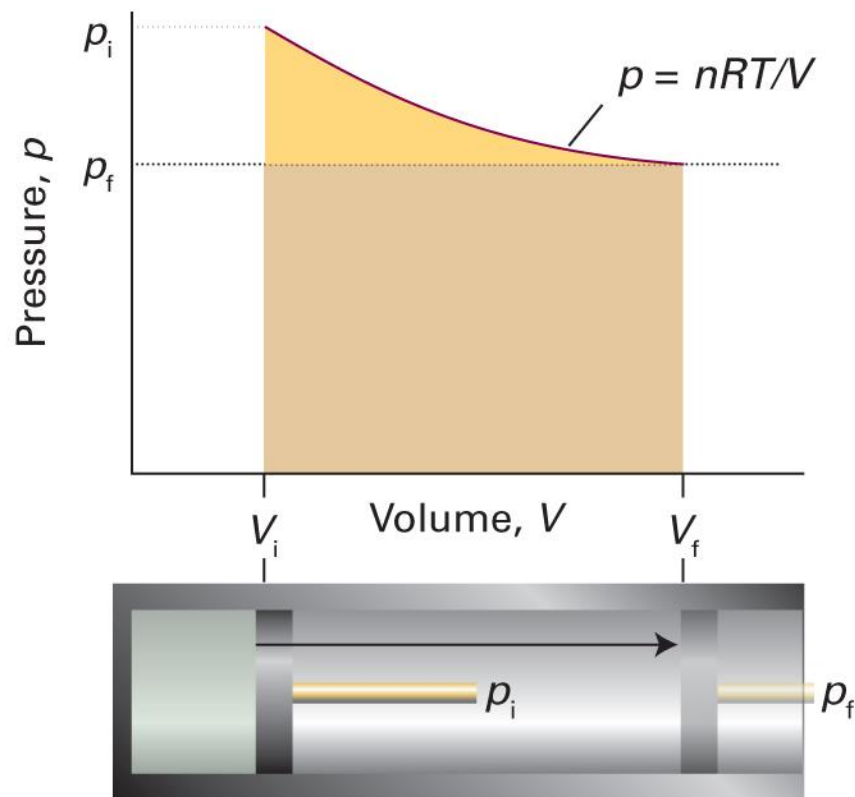
Perfect Gas, Constant T

For an **isothermal** reversible expansion,

$$w = -nR \int_{V_i}^{V_f} \frac{T}{V} dV = -nRT \int_{V_i}^{V_f} \frac{1}{V} dV$$

$$w = -nRT (\ln V_f - \ln V_i) = -nRT \ln \frac{V_f}{V_i}$$

Perfect Gas, Constant T



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2A.7)

Brief Illustration 2A.3

When a sample of 1.00 mol Ar, regarded here as a perfect gas, undergoes an isothermal reversible expansion at 20.0 °C from 10.0 dm³ to 30.0 dm³, what is the work done?

$$\begin{aligned} w &= -nRT \ln(V_f/V_i) \\ &= -(1.00 \text{ mol}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (293 \text{ K}) \times \ln(30.0/10.0) \\ &= -2.68 \times 10^3 \text{ J} \end{aligned}$$

Heat

$$q \propto \Delta T$$

For infinitesimal dT ,

$$\delta q \propto dT$$

$$\delta q = C dT$$

(C : heat capacity)

Calculation of Heat

$$\delta q = C dT$$

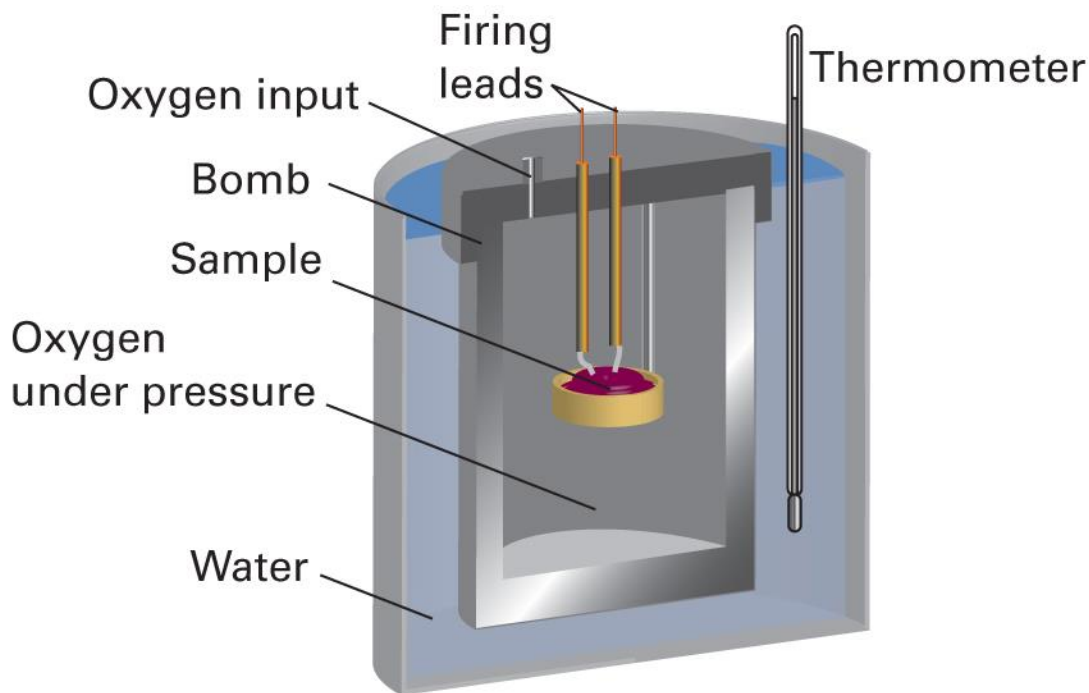
- Total heat for a change from **state i** to **state f**

$$q = \int_{\text{state } i}^{\text{state } f} \delta q = \int_{\text{state } i}^{\text{state } f} (C dT) = \int_{T_i}^{T_f} C dT$$

Heat Capacity

- Molar heat capacity $C_m = C/n$
- Specific heat capacity $C_s = C/m$
- Heat capacity depends on the constraint
 - C_V : heat capacity at constant volume ($C_{V,m}$, $C_{V,s}$)
 - C_p : heat capacity at constant pressure ($C_{p,m}$, $C_{p,s}$)

Calorimetry: Heat Measurement



First Law Revisited

$$dU = \delta q + \delta w$$

$$dU = \delta q + \delta w_{\text{exp}} + \delta w_{\text{add}}$$

$$dU = C dT - p_{\text{ex}} dV + \delta w_{\text{add}}$$

For $\delta w_{\text{add}} = 0$,

$$dU = C dT - p_{\text{ex}} dV$$

At Constant V

$$dU = C dT - p_{\text{ex}} dV$$

At constant volume, $dV = 0$ and $C = C_V$

$$dU = C_V dT$$

The change in internal energy is

$$\Delta U = \int_{\text{state } i}^{\text{state } f} dU = \int_{\text{state } i}^{\text{state } f} (C_V dT) = \int_{T_i}^{T_f} C_V dT$$

At Constant V

$$\Delta U = \int_{T_i}^{T_f} C_V dT$$

If C_V is constant,

$$\Delta U = \int_{T_i}^{T_f} C_V dT = C_V \int_{T_i}^{T_f} dT = C_V(T_f - T_i) = C_V\Delta T$$

Since $\Delta U = q + w = q$ at constant volume,

$$q = C_V\Delta T$$

Not in Atkins

(Monatomic) Perfect Gas

$$U = \frac{3}{2}nRT$$

$$dU = \frac{3}{2}nRdT$$

$$dU = CdT - p_{\text{ex}}dV$$

$$\frac{3}{2}nRdT = CdT - p_{\text{ex}}dV$$

Not in Atkins

Perfect Gas at Constant V

$$\frac{3}{2}nRdT = CdT - p_{\text{ex}}dV$$

At constant volume, $dV = 0$ and $C = C_V$

$$\frac{3}{2}nRdT = C_VdT$$

Hence,

$$C_V = \frac{3}{2}nR$$